CLINICAL · NOSE & SINUSES

# Nasal foreign bodies, rhinoliths and myiasis

A child with a one-sided, foul-smelling, sometimes bloodstained nasal discharge has a foreign body in that nostril until proven otherwise. Most are harmless beads and seeds; a button battery is an emergency. In adults the same story may be a rhinolith built around a forgotten object, and in a neglected, foul nose flies may lay their eggs.

**AT A GLANCE**

- One-sided foul discharge in a child: foreign body.
- Button batteries and magnets: remove urgently.
- Rhinolith: a stone around an old foreign body.
- Maggots: kill them first, then remove; look for the cause.

## 1 CHAPTER ONE Overview

Things that should not be in the nose.

### 1.1 The conditions

- Foreign bodies, mostly in children.
- Rhinoliths: stones formed around a nucleus in the nose.
- Nasal myiasis: maggots in the nose.
- Nasal synechiae: adhesions, often the aftermath of injury or surgery.

### 1.2 The dictum

*If a child presents with unilateral, foul-smelling nasal discharge, a foreign body must be excluded.* Other causes of one-sided bloodstained discharge in a child are rhinolith, nasal diphtheria, myiasis, sinusitis and, rarely, a tumour.

CHAPTER TWO

# 2 Foreign bodies

Mostly children.

## 2.1 Types

- **Inorganic:** beads, buttons, pebbles, chalk, pieces of plastic and toys; non-irritant, may stay silent for a long time.
- **Organic:** seeds, peas, beans, nuts, paper and sponge; they absorb water and swell, and cause early inflammation.
- **Dangerous:** button (disc) batteries leak alkali and produce electrical currents that destroy mucosa within hours, causing septal perforation; paired magnets on either side of the septum cause pressure necrosis.
- **Iatrogenic:** cotton pledgets or swabs left behind.

## 2.2 Clinical picture

A history of insertion if it was seen; otherwise one-sided obstruction and foul, sometimes bloodstained discharge with excoriation of the nostril. The object often lies on the floor or between the septum and inferior turbinate, buried in granulation and discharge.

## 2.3 Investigations

Usually none. A radiograph helps with metallic or radio-opaque objects and button batteries (a double ring on the radiograph).

## 2.4 Removal

- **Soft objects** (paper, cotton): forceps under good light.
- **Round, smooth objects:** a blunt hook or a Eustachian catheter passed beyond the object and drawn forwards along the floor; never forceps, which push it deeper.
- **Positive pressure:** in a cooperative child, a parent blowing into the mouth while the clear nostril is blocked (*mother's kiss*), or a bag-valve device.

- **Under general anaesthesia:** for uncooperative children and posterior or impacted objects, with a cuffed tube, the head extended (Rose's position) and a pack in the nasopharynx to prevent inhalation; an endoscope helps to find and remove it.

**BUTTON BATTERIES**

Remove immediately. Do not instil drops or water, which speed the electrical burn. Check afterwards for septal necrosis and perforation.

### 2.5 Complications

Rhinitis and sinusitis, epistaxis, septal perforation, rhinolith formation, and inhalation into the airway during removal.

## 3 CHAPTER THREE Rhinolith

A stone in the nose.

### 3.1 Formation

Slow deposition of calcium and magnesium salts around a nucleus: a small foreign body (*exogenous*), or a blood clot or dried secretions (*endogenous*). Over years it grows into a large, irregular mass that can erode the septum or lateral wall.

### 3.2 Clinical picture

More common in adults: one-sided obstruction with a foul, often bloodstained discharge, sometimes epistaxis or neuralgic pain. A grey-brown or greenish-black, irregular, stony-hard and brittle mass lies between the septum and turbinates, often surrounded by granulation. A radiograph or CT shows the calcified mass.

### 3.3 Treatment

Removal under general anaesthesia, usually through the nostril, breaking large

stones into pieces; rarely a lateral rhinotomy for very large, hard stones.

# 4 CHAPTER FOUR Nasal myiasis

Maggots in the nose.

## 4.1 How it happens

Flies, especially *Chrysomya* species, are attracted by the foul discharge of atrophic rhinitis, syphilis, leprosy, rhinosporidiosis or infected wounds. They lay about 200 eggs at a time, which hatch into larvae within a day. In India it is seen mostly from August to October.

## 4.2 Clinical picture

For the first three or four days: intense irritation, sneezing, watering eyes, headache and a thin bloodstained discharge, often presenting as epistaxis. The lids and lips swell; then maggots crawl out of the nose, and the patient smells foul. They burrow and destroy the nose, sinuses, face, palate and eye, form fistulae, and can cause meningitis and death.

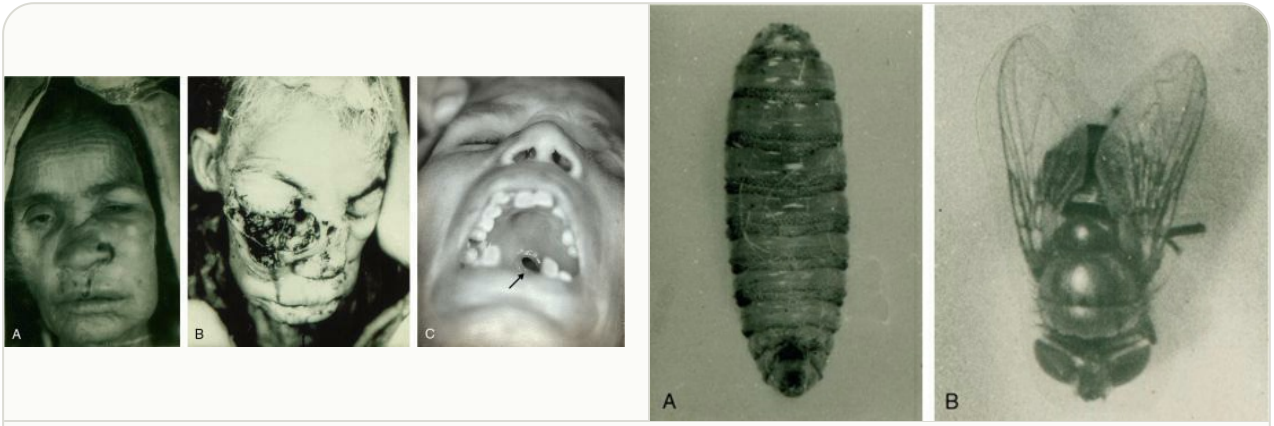



**FIGURE 1** First, nasal myiasis: swelling of the nose and lids with bloodstained discharge, destruction of the cheek and eye in a neglected case, and a palatal perforation (arrow). Second, the maggot and the fly that lays it.

Dhingra, figs 29.1 and 29.2

### 4.3 Treatment

- Pick out every visible maggot with forceps; they retreat from light into dark recesses.
- Instil chloroform water (or turpentine, or ivermectin) to kill or suffocate them, and remove them over several sessions.
- Warm saline douches to clear slough and dead maggots; systemic ivermectin and antibiotics.
- Isolate the patient under a net, away from flies.
- Endoscopy and imaging to find hidden maggots, and treat the underlying disease.

## 5 CHAPTER FIVE Nasal synechiae

Adhesions.

### 5.1 Cause

Scar bands between the septum and lateral wall where two raw mucosal surfaces touch: after septal or sinus surgery, polypectomy, removal of foreign bodies, reduction of fractures, cautery, packing, or ulcerating infections.

### 5.2 Effects

Nasal obstruction, and blocked sinus drainage with sinusitis, headache and discharge.

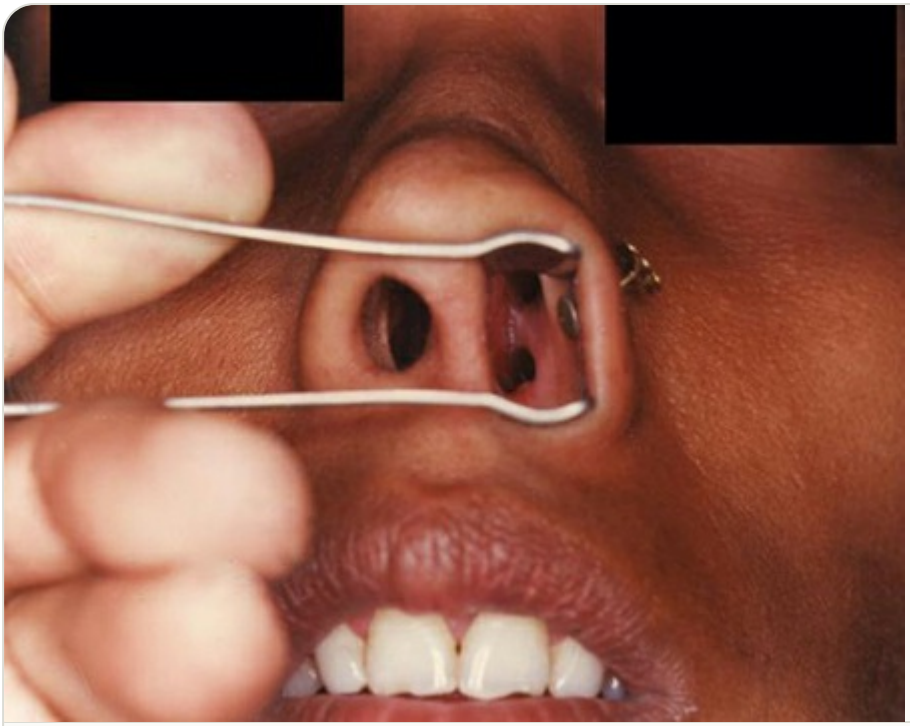


**FIGURE 2** A synechia in the left nasal cavity.
Dhingra, fig. 29.3

### 5.3 Treatment

Divide the adhesion and keep the raw surfaces apart with a thin silastic sheet (splint) until healed, about one to three weeks. Prevention: avoid opposing raw areas and use splints after septal and turbinate surgery.

## 6 CHAPTER SIX For the examination

What the examiner wants to hear.

### 6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The dictum:** one-sided foul discharge in a child.
- **Removal technique** for round objects, and why not forceps.
- **Button battery** as an emergency.
- **Rhinolith:** formation and appearance.

- **Myiasis:** predisposing diseases and treatment.

## 6.2 Common pitfalls

COMMON PITFALLS

- Treating one-sided discharge in a child with antibiotics without looking.
- Grasping a bead with forceps and pushing it backwards.
- Putting drops into a nose with a button battery.

Written for this app from Oxford Textbook of Otolaryngology ch. 77, 78, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 25, Dhingra ch. 29 and K. J. Lee ch. 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

## Foreign Bodies

### IMAGING FINDINGS

- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

### TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

### CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…

NOSE & NASAL CAVITY · CONGENITAL

# Choanal Atresia

## IMAGING FINDINGS

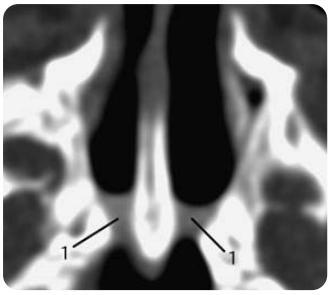
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

## TOP DIFFERENTIALS

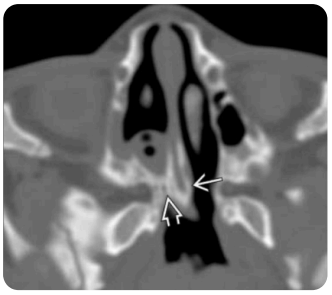
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

## CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears




CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.